

Galera Cluster SOP (Standard Operating Procedure)

1. Introduction

Galera Cluster is a synchronous multi-master database cluster for MySQL/MariaDB.

2. Key Concepts

Multi-master replication, quorum, SST, IST, certification-based replication.

3. Architecture

Minimum 3 nodes recommended. 2 nodes require garbd arbitrator.

4. Installation (Ubuntu 24.04)

```
apt update
apt install mariadb-server -y
```

5. Configuration

```
[galera]
wsrep_on=ON
wsrep_provider=/usr/lib/galera/libgalera_smm.so
wsrep_cluster_name="galera_lab"
wsrep_cluster_address="gcomm://IP1,IP2,IP3"
wsrep_node_name="node1"
wsrep_node_address="IP1"
wsrep_sst_method=rsync
binlog_format=ROW
default_storage_engine=InnoDB
innodb_autoinc_lock_mode=2
bind-address=0.0.0.0
```

6. Bootstrap Cluster

```
systemctl stop mariadb
galera_new_cluster
```

7. Add Nodes

```
systemctl start mariadb
```

8. Verification

```
SHOW STATUS LIKE 'wsrep_cluster_size';
SHOW STATUS LIKE 'wsrep_ready';
```

9. Testing

```
CREATE DATABASE test;  
CREATE TABLE test.users(id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50));  
INSERT INTO test.users(name) VALUES('test');
```

10. Failure Handling

Check grastate.dat and bootstrap correct node.

11. Split Brain Handling

Use quorum rules. Never bootstrap multiple nodes.

12. 2 Node + garbd Setup

Install galera-arbitrator and configure /etc/default/garb.

13. Monitoring

```
SHOW STATUS LIKE 'wsrep%';
```

14. Best Practices

Use 3 nodes, low latency network, backups, HAProxy.

15. Conclusion

Galera provides strong consistency and high availability.